MATH 181A - Homework 8

due Thursday, March 5 at 11:59PM

Problem 1. 7.4.12

If a normally distributed sample of size $n = 16$ produces a 95% confidence interval for μ that ranges from 44.7 to 49.9, what are the values of $\bar{y}$ and s?

Solution: Since the interval is symmetric about $\bar{y}$, we have $\bar{y} = \frac{44.7+49.9}{2} = 47.3$. For $n = 16$ with unknown σ, we use the t-distribution: $t_{0.025,15} \approx 2.131$. The radius of the interval is given by

$$2.131\frac{s}{\sqrt{16}} = \frac{49.9 - 44.7}{2} = 2.6 \implies s = \frac{2.6 \cdot 4}{2.131} \approx 4.88$$

Problem 2. 7.4.20

Make the decision in two ways using the t-table in Appendix A.2: (1) finding the critical region; and (2) finding the range of the p-value.

Note: the sample mean and sample standard deviation of the 34 ratios in the table are 0.6373 and 0.14139 respectively.

In addition to the Shoshoni data of Case Study 7.4.2, a set of rectangles that might tend to the golden ratio are national flags. The table below gives the width-to-length ratios for a random sample of the flags of thirty-four countries. Let μ be the width-to-length ratio for national flags. At the $\alpha = 0.01$ level, test $H_0 : \mu = 0.618$ versus $H_1 : \mu \neq 0.618$.

Country	Ratio Width to Height	Country	Ratio Width to Height
Afghanistan	0.500	Iceland	0.720
Albania	0.714	Iran	0.571
Algeria	0.667	Israel	0.727
Angola	0.667	Laos	0.667
Argentina	0.667	Lebanon	0.667
Bahamas	0.500	Liberia	0.526
Denmark	0.757	Macedonia	0.500
Djibouti	0.553	Mexico	0.571
Ecuador	0.500	Monaco	0.800
Egypt	0.667	Namibia	0.667
El Salvador	0.600	Nepal	1.250
Estonia	0.667	Romania	0.667
Ethiopia	0.500	Rwanda	0.667
Gabon	0.750	South Africa	0.667
Fiji	0.500	St. Helena	0.500
France	0.667	Sweden	0.625
Honduras	0.500	United Kingdom	0.500

Data from: http://www.anyflag.com/country/costaric.php

(a) Finding the critical region.

Solution: The critical region is when $|t| > t_{0.005}$ where $t_{0.005}$ is found from the (df=33, 0.005) entry of the t-table. We find that $t_{0.005} = 2.7333$. Hence, the critical region is when $|t| > 2.7333$. Our t-statistic is given by

$$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}} = \frac{0.6373 - 0.618}{0.14139/\sqrt{34}} = 0.7959$$

Since $|0.7959| < 2.7333$, we fail to reject H_0.

(b) Finding the range of the p-value.

Solution: Using our previously calculated t-statistic, we can see in the t-table that a t-statistic of 0.7959 has greater p-value than 0.20 (the largest p-value in the table) which has a statistic of 0.8527. Hence, $p > 2 \cdot 0.20 = 0.40 > \alpha = 0.01$, so we fail to reject H_0.

Problem 3. 7.4.21

Make the decision in two ways using the t-table in Appendix A.2: (1) finding the critical region; and (2) finding the range of the p-value.

A manufacturer of pipe for laying underground electrical cables is concerned about the pipe's rate of corrosion and whether a special coating may retard that rate. As a way of measuring corrosion, the manufacturer examines a short length of pipe and records the depth of the maximum pit. The manufacturer's tests have shown that in a year's time in the particular kind of soil the manufacturer must deal with, the average depth of the maximum pit in a foot of pipe is 0.0042 inch. To see whether that average can be reduced, ten pipes are coated with a new plastic and buried in the same soil. After one year, the following maximum pit depths are recorded (in inches): 0.0039, 0.0041, 0.0038, 0.0044, 0.0040, 0.0036, 0.0034, 0.0036, 0.0046, and 0.0036. Given that the sample standard deviation for these ten measurements is 0.000383 inch, can it be concluded at the $\alpha = 0.05$ level of significance that the plastic coating is beneficial?

(a) Finding the critical region.

Solution: The critical region is when $t < -t_{0.05}$ where $t_{0.05}$ is found from the (df=9, 0.05) entry of the t-table. We find that $t_{0.05} = 1.8331$. Hence, the critical region is when $t < -1.8331$. Our t-statistic is given by

$$t = \frac{\bar{y} - \mu_0}{s/\sqrt{n}} = \frac{0.0039 - 0.0042}{0.000383/\sqrt{10}} = -2.4770$$

Since $-2.4770 < -1.8331$, we reject H_0.

(b) Finding the range of the p-value.

Solution: Using our previously calculated t-statistic, we can see in the t-table that a t-statistic of $-2.4770 \in (2.2622, 2.821)$ which are p-values of 0.025 and 0.01 respectively. Hence, $p \in (0.01, 0.025) < \alpha = 0.05$, so we reject H_0.

Problem 4. 7.5.8

A random sample of size $n = 19$ is drawn from a normal distribution for which $\sigma^2 = 12.0$. In what range are we likely to find the sample variance, s^2? Answer the question by finding two numbers a and b such that

$$P(a \leq S^2 \leq b) = 0.95$$

Solution: We use the fact that $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. With $n = 19$ and $\sigma^2 = 12$, we have $\frac{18S^2}{12} \sim \chi^2_{18}$. For a 95% central interval, we need $P(\chi^2_{0.025,18} \leq \frac{18S^2}{12} \leq \chi^2_{0.975,18}) = 0.95$. From the chi-squared table: $\chi^2_{0.025,18} \approx 8.23$ and $\chi^2_{0.975,18} \approx 31.53$. Hence

$$a = \frac{8.23 \cdot 12}{18} \approx 5.49, \qquad b = \frac{31.53 \cdot 12}{18} \approx 21.02$$

Problem 5. 7.3.4

Use the fact that $(n-1)S^2/\sigma^2$ is a chi square random variable with $n-1$ df to prove that

$$\text{Var}(S^2) = 2\sigma^4/(n-1)$$

(*Hint:* Use the fact that the variance of a chi square random variable with k df is $2k$.)

Solution: We use the fact that $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. The variance of a chi-squared random variable with k df is $2k$. Hence

$$\text{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) = \left(\frac{n-1}{\sigma^2}\right)^2 \text{Var}(S^2) = 2(n-1) \implies \text{Var}(S^2) = \frac{2\sigma^4}{n-1}$$

Problem 6. 7.3.10

Suppose $X_1, \ldots, X_n$ are i.i.d. random variables having a $N(\mu, \sigma^2)$ distribution. Consider the estimator of σ^2,

$$\hat{\sigma}_c^2 = c\sum_{i=1}^{n}(X_i - \bar{X})^2$$

Recall that $\hat{\sigma}_c^2$ is the MLE if $c = 1/n$ and is the unbiased estimator S^2 if $c = 1/(n-1)$.

(a) Calculate the mean squared error of $\hat{\sigma}_c^2$.

Solution: We have $\hat{\sigma}_c^2 = c\sum_{i=1}^{n}(X_i - \bar{X})^2 = c(n-1)S^2$. We know that $\mathbb{E}[S^2] = \sigma^2$ and $\text{Var}(S^2) = 2\sigma^4/(n-1)$. We also know:

$$\text{MSE}(\hat{\sigma}_c^2) = \text{Var}(\hat{\sigma}_c^2) + \big(\mathbb{E}[\hat{\sigma}_c^2] - \sigma^2\big)^2.$$

$\mathbb{E}[\hat{\sigma}_c^2] = c(n-1)\mathbb{E}[S^2] = c(n-1)\sigma^2$, so Bias $= c(n-1)\sigma^2 - \sigma^2 = \sigma^2\big(c(n-1)-1\big)$.

$\text{Var}(\hat{\sigma}_c^2) = c^2(n-1)^2\text{Var}(S^2) = c^2(n-1)^2 \cdot \frac{2\sigma^4}{n-1} = 2c^2(n-1)\sigma^4$.

$$\text{MSE}(\hat{\sigma}_c^2) = 2c^2(n-1)\sigma^4 + \sigma^4\big(c(n-1)-1\big)^2.$$

(b) For which value of c is the mean squared error the smallest?

Solution: We find the minimum MSE by taking the derivative of the MSE with respect to c and setting it to 0.

$$\begin{aligned}\frac{d}{dc}\text{MSE}(\hat{\sigma}_c^2) = 4c(n-1)\sigma^4 + 2(n-1)\sigma^4(c(n-1)-1) &= 0\\ \implies 2c + nc - c - 1 &= 0\\ \implies c &= \frac{1}{n+1}\end{aligned}$$

Problem 7. R Simulation

Use 5000 repetitions to generate two empirical density functions in the same plot to verify

(a) The empirical distribution of $X_1^2 + X_2^2 + \cdots + X_9^2$ (using a solid line) is close to that of χ_9^2 (using a dashed line), where $X_i \sim$ i.i.d.$N(0, 1)$.

```
set.seed(42)
n_rep <- 5000

png("7a.png", width = 8, height = 5, units = "in", res = 150)  # landscape

empirical <- replicate(n_rep, sum(rnorm(9, 0, 1)^2))

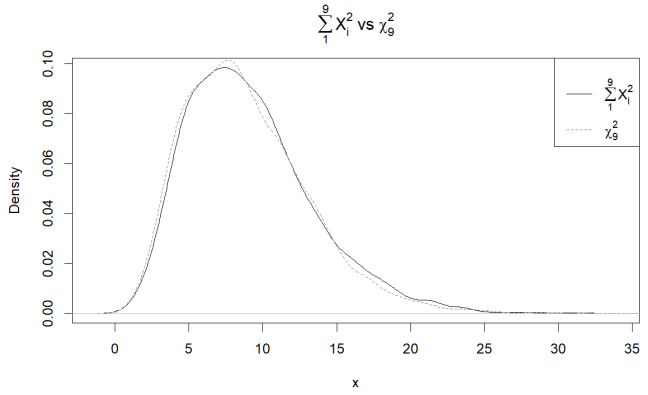
theoretical <- rchisq(n_rep, df = 9)

dens_emp <- density(empirical)
dens_theo <- density(theoretical)

plot(dens_emp, type = "l", lty = 1, col = "black",
        xlab = "x", ylab = "Density",
        main = expression(sum(X[i]^2, i = 1, 9) ~ "vs" ~ chi[9]^2))
lines(dens_theo, lty = 2, col = "gray50")
legend("topright",
        legend = c(expression(sum(X[i]^2, i = 1, 9)), expression(chi[9]^2)),
        lty = c(1, 2), col = c("black", "gray50"))

dev.off()
```

Solution:



(b) The empirical distribution of $3\bar{X}/S$ (using a solid line) is close to that of t_8 (using a dashed line), where $X_i \sim$ i.i.d.$N(0,1)$, $\bar{X} = \frac{1}{9}\sum_{i=1}^{9} X_i$, and $S^2 = \frac{1}{8}\sum_{i=1}^{9}(X_i - \bar{X})^2$.

```
set.seed(42)
n_rep <- 5000

png("7b.png", width = 8, height = 5, units = "in", res = 150)  # landscape

empirical <- replicate(n_rep, {
  x <- rnorm(9, 0, 1)
  3 * mean(x) / sqrt((1 / 8) * sum((x - mean(x))^2))
})

theoretical <- rt(n_rep, df = 8)

dens_emp <- density(empirical)
dens_theo <- density(theoretical)

plot(dens_emp, type = "l", lty = 1, col = "black",
        xlab = "x", ylab = "Density",
        main = expression(3 * bar(X) / S ~ "vs" ~ t[8]))
lines(dens_theo, lty = 2, col = "gray50")
legend("topright",
        legend = c(expression(3 * bar(X) / S), expression(t[8])),
        lty = c(1, 2), col = c("black", "gray50"))

dev.off()
```

Solution:

